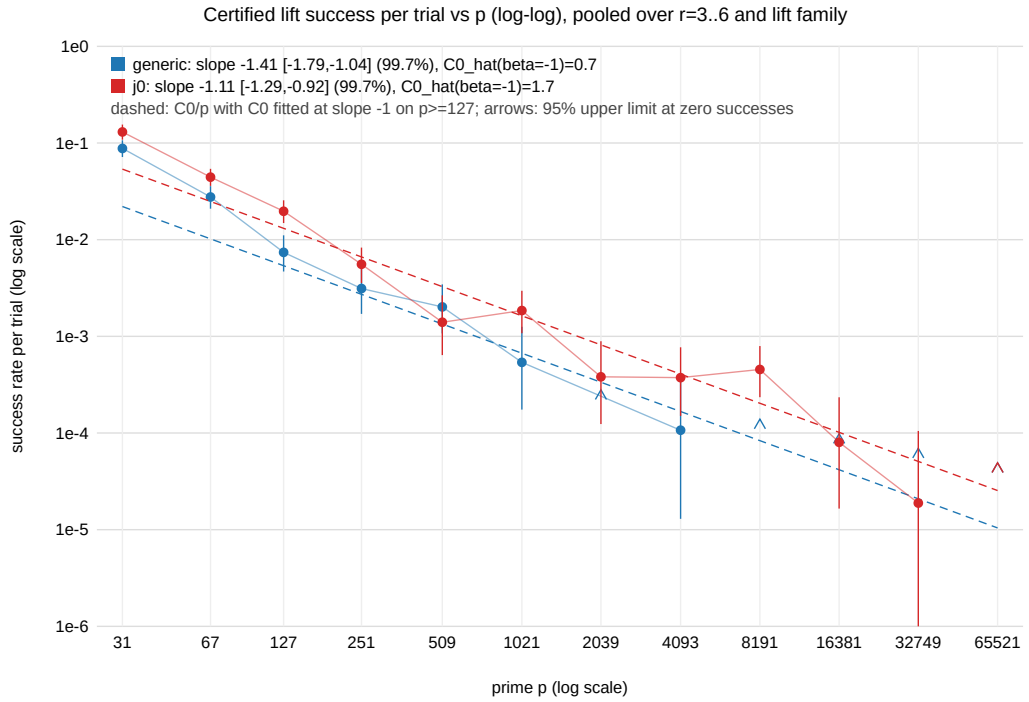
Xedni lift success sweep

Xedni lift success per trial over toy primes: results (2026-11-12)



Lift success vs p

Protocol: PROTOCOL.md, frozen by sha256 in `FREEZE.json` before any row existed. All numbers below are `[pcc:unverified]`: no pcc claim has been filed, and this session produced them (sabotage is assumed; see `AGENTS.md`). Raw rows: `data/rows-p*.jsonl.gz`; analysis `analyze.py` -> `results.json`; graph `success_vs_p.svg`; log `run.log`.

Verdict

- The frozen prediction is **supported by its decision rule**: no unit is classed LEAD, and the three pooled units are in class “tracks 1/p” (`pool/j0`) or “faster than 1/p” (`pool/all`, `pool/generic`). No family of the 16 is bounded away from `1/p`. `[pcc:unverified]`
- One part of the prediction text is **not met as worded**: the prediction asked the pooled slope point estimates to lie in [-1.25, -0.75]. Measured (window `p >= 127`): `pool/all` -1.174 (99.7% interval -1.337 to -1.012), `pool/j0` -1.105 (-1.287 to -0.924), `pool/generic` -1.412 (-1.788 to -1.037). `pool/generic` lies outside [-1.25, -0.75]; its interval is entirely below -1, i.e. it falls faster than `1/p`, which the bound allows.
- Fitted constants with the slope fixed at -1 (`C0_hat`, window `p >= 127`): `pool/all` 1.16, `pool/generic` 0.68, `pool/j0` 1.66 (`p x rate` at `p = 127`: 1.68, 0.94, 2.50). At `p =`

65521: 0 successes in 150,057 trials, 95% upper limit 2.46e-05 per trial, i.e. `p x rate <= 1.61`.

- **Most successes are not deep relations.** Of 572 certified successes, 476 (83%) are *line* events (three lifted points collinear, a relation `+/-P_i +/- P_j +/- P_k = 0` with unit coefficients), 3 are single-point torsion relations, and 96 are non-line. Non-line successes: 59 at p=31, 24 at 67, 9 at 127, 2 at 251, 1 at 509, 1 at 4093, none above (see the class column; they carry no slope claim (every non-line unit is "uninformative"; the pooled non-line count at `p >= 127` is 13). `dl_recovered` holds for 423 of 572 successes (the relation had `n_Q` coprime to the point order and gave `l`).
- Nothing here is an improvement over the index-calculus baseline, a relabelling or accounting change, or a statement about ECDLP cost against Pollard rho (not computed). Distance remaining to matched rho: not measured by this experiment.

What ran

item	value
base commit	`0d69fc5526c02c4071db896d8800be74929f4ff7`
`ca-icx` sha256	`53addfe29282034d2715b33e65edd686b24a2e79e9a20e91dd1c3`
trials scheduled / ok	504,304 / 500,242
excluded (reported, not in denominators)	small_order 3,780; no_integer_solution 280; rank_deficient 2
certificate failures (`lift_cert_failed`, `relation_cert_failed`, `lift_error`)	0
wall time	2152 s on 4 processes, all 12 prime stages ran (none censored)
detector selftest	all 6 checks pass (`python3 sweep.py selftest`)
waiver	S3 checkpoint waived by the owner for this toy-size run (2026-11-PROTOCOL.md

Each success was verified exactly: lifted curve non-singular over Q, reduces to E mod p, lifted points on it and reducing to the intended points, and the relation summing to O over Q by exact rational arithmetic (twice), and holding in E(F_p). [pcc:unverified]

Success counts per prime (pooled over r = 3..6 and both lift families)

pool/all

p	trials (ok)	successes	rate	95% CI (Garwood)	p x rate	non-line	recovers l	F_p-box relation
31	2114	226	1.069e-01	[9.34e-02, 1.22e-01]	3.31	59	170	2114
67	4273	155	3.627e-02	[3.08e-02, 4.25e-02]	2.43	24	111	4273
127	5959	79	1.326e-02	[1.05e-02, 1.65e-02]	1.68	9	56	5959
251	8796	38	4.320e-03	[3.06e-03, 5.93e-03]	1.08	2	28	8796

p	trials (ok)	successes	rate	95% CI (Garwood)	p x rate	non-line	recovers l	F_p-box relation
509	12884	22	1.708e-03	[1.07e-03, 2.59e-03]	0.87	1	13	12859
1021	18487	22	1.190e-03	[7.46e-04, 1.80e-03]	1.22	0	20	17704
2039	26293	5	1.902e-04	[6.17e-05, 4.44e-04]	0.39	0	4	24502
4093	37434	9	2.404e-04	[1.10e-04, 4.56e-04]	0.98	1	7	32585
8191	52917	12	2.268e-04	[1.17e-04, 3.96e-04]	1.86	0	11	43567
16381	74939	3	4.003e-05	[8.26e-06, 1.17e-04]	0.66	0	2	58428
32749	106089	1	9.426e-06	[2.39e-07, 5.25e-05]	0.31	0	1	72650
65521	150057	0	0.000e+00	[0.00e+00, 2.46e-05]	0.00	0	0	101364

pool/generic

p	trials (ok)	successes	rate	95% CI (Garwood)	p x rate	non-line	recovers l	F_p-box relation
31	1161	102	8.786e-02	[7.16e-02, 1.07e-01]	2.72	32	79	1161
67	2063	57	2.763e-02	[2.09e-02, 3.58e-02]	1.85	7	39	2063
127	3112	23	7.391e-03	[4.69e-03, 1.11e-02]	0.94	1	15	3112
251	4483	14	3.123e-03	[1.71e-03, 5.24e-03]	0.78	1	10	4483
509	6451	13	2.015e-03	[1.07e-03, 3.45e-03]	1.03	1	7	6430
1021	9292	5	5.381e-04	[1.75e-04, 1.26e-03]	0.55	0	5	8910
2039	13179	0	0.000e+00	[0.00e+00, 2.80e-04]	0.00	0	0	11990
4093	18709	2	1.069e-04	[1.29e-05, 3.86e-04]	0.44	1	2	16304
8191	26501	0	0.000e+00	[0.00e+00, 1.39e-04]	0.00	0	0	21240
16381	37498	0	0.000e+00	[0.00e+00, 9.84e-05]	0.00	0	0	27835
32749	53041	0	0.000e+00	[0.00e+00, 6.95e-05]	0.00	0	0	35449
65521	75044	0	0.000e+00	[0.00e+00, 4.92e-05]	0.00	0	0	47881

pool/j0

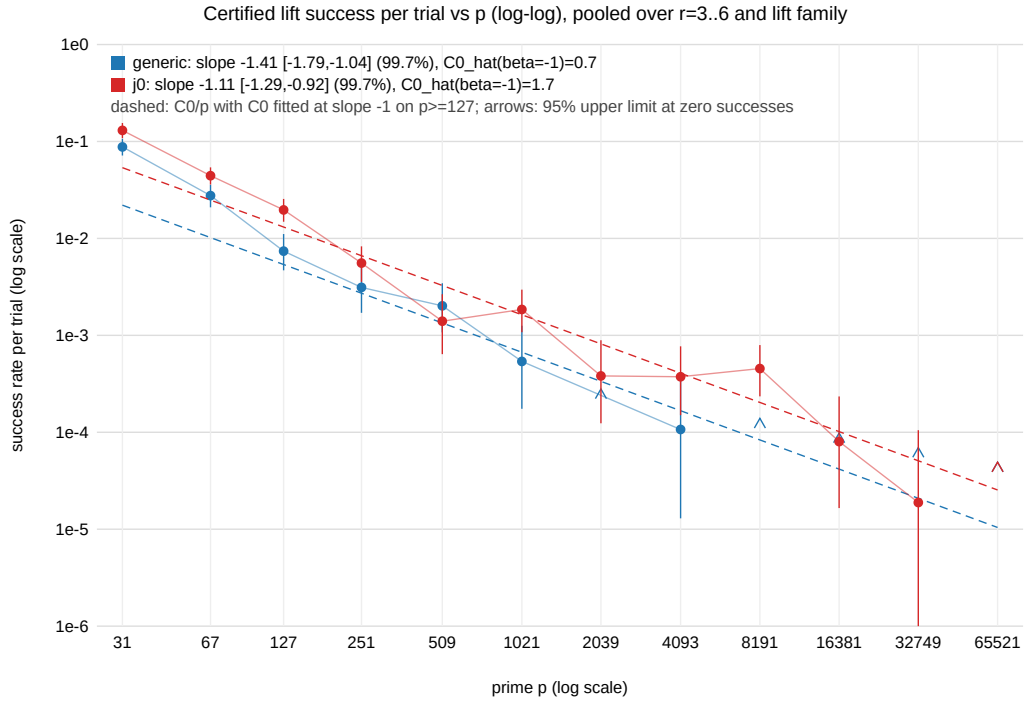
p	trials (ok)	successes	rate	95% CI (Garwood)	p x rate	non-line	recovers l	F_p-box relation
31	953	124	1.301e-01	[1.08e-01, 1.55e-01]	4.03	27	91	953
67	2210	98	4.434e-02	[3.60e-02, 5.40e-02]	2.97	17	72	2210
127	2847	56	1.967e-02	[1.49e-02, 2.55e-02]	2.50	8	41	2847
251	4313	24	5.565e-03	[3.57e-03, 8.28e-03]	1.40	1	18	4313
509	6433	9	1.399e-03	[6.40e-04, 2.66e-03]	0.71	0	6	6429
1021	9195	17	1.849e-03	[1.08e-03, 2.96e-03]	1.89	0	15	8794
2039	13114	5	3.813e-04	[1.24e-04, 8.90e-04]	0.78	0	4	12512
4093	18725	7	3.738e-04	[1.50e-04, 7.70e-04]	1.53	0	5	16281
8191	26416	12	4.543e-04	[2.35e-04, 7.94e-04]	3.72	0	11	22327
16381	37441	3	8.013e-05	[1.65e-05, 2.34e-04]	1.31	0	2	30593
32749	53048	1	1.885e-05	[4.77e-07, 1.05e-04]	0.62	0	1	37201
65521	75013	0	0.000e+00	[0.00e+00, 4.92e-05]	0.00	0	0	53483

Units

unit (curve / lift / r)	successes p>=127	beta	99.7% interval	C0_hat	class	non-line succ. (p>=127)
generic/balanced/3	13	-1.91	[-3.11, -0.71]	1.24	tracks 1/p	1
generic/balanced/4	8	n/a	n/a	n/a	uninformative	2
generic/balanced/5	12	-1.45	[-2.31, -0.60]	1.15	tracks 1/p	0
generic/balanced/6	16	-1.35	[-2.01, -0.68]	1.55	tracks 1/p	0
generic/wide/3	3	n/a	n/a	n/a	uninformative	0
generic/wide/4	4	n/a	n/a	n/a	uninformative	1
generic/wide/5	0	n/a	n/a	n/a	uninformative	0
generic/wide/6	1	n/a	n/a	n/a	uninformative	0
j0/balanced/3	36	-1.35	[-1.79, -0.90]	3.44	tracks 1/p	3
j0/balanced/4	20	-1.22	[-1.75, -0.70]	1.98	tracks 1/p	2

unit (curve / lift / r)	successes p>=127	beta	99.7% interval	C0_hat	class	non-line succ. (p>=127)
j0/balanced/5	23	-0.92	[-1.29, -0.55]	2.31	tracks 1/p	0
j0/balanced/6	28	-1.02	[-1.39, -0.66]	2.90	tracks 1/p	2
j0/wide/3	17	-1.16	[-1.70, -0.62]	1.63	tracks 1/p	2
j0/wide/4	2	n/a	n/a	n/a	uninformative	0
j0/wide/5	1	n/a	n/a	n/a	uninformative	0
j0/wide/6	7	n/a	n/a	n/a	uninformative	0
pool/all	191	-1.17	[-1.34, -1.01]	1.16	faster than 1/p	13
pool/generic	57	-1.41	[-1.79, -1.04]	0.68	faster than 1/p	4
pool/j0	134	-1.11	[-1.29, -0.92]	1.66	tracks 1/p	9

The last column is the number of ok trials in which some non-zero vector of the box satisfies the relation in E(F_p), a necessary condition for success; it is close to all trials at every p, so the success events are rare inside a set of instances that nearly all have a mod-p relation.



success rate vs p

Checks beyond the preregistered analysis (post hoc, labelled)

- Pearson dispersion of the pooled Poisson fits (window `p >= 127`): `pool/all` 2.87, `pool/generic` 1.04, `pool/j0` 4.05 (`dispersion.json`). Trials within a cell share 40 curves, and `pool/j0` at p=8191 has 12 successes (about 5 expected from a slope-minus-one fit), 8 of them on three of its 40 curves. The preregistered intervals assume Poisson counts and are therefore too narrow for `pool/all` and `pool/j0`. Inflating the standard errors by the root of the dispersion widens `pool/all` to about [-1.45, -0.90],

which contains -1 (class “tracks 1/p” instead of “faster than 1/p”); `pool/j0` stays “tracks”. Neither change affects the verdict.
- The p = 31 and p = 67 cells (outside the window) have `p x rate` of 2.4 to 4.0, above the later values: the rate is saturating there, as expected when the group order is of the size of the box. [pcc:unverified]

Scorecard

check	**outcome**
frozen decision rule: any LEAD unit	none (0 of 19 units)
pooled units in class 2 or 3	yes (3 of 3)
prediction text: pooled slopes in [-1.25, -0.75]	2 of 3 (`pool/generic` -1.41)
every success certified	572 of 572
controls (planted relation found, mutation rejected, negative pair, primality)	pass
uninformative units (< 10 successes at `p >= 127`)	8 of 16
non-line successes at `p >= 127`	13 (pooled); no slope claim possible
claim ids	none filed; everything `[pcc:unverified]`

Limits

- The relation box is |n_i| <= 6; dependent lifts with larger relations count as failures, so rates are lower bounds on dependence.
- `r = 3..6`, seven-coefficient path only; lifts are the repository optimizer’s (LLL + Babai + radius 1), balanced or balanced plus a multiple of p. Toy primes only (31 to 65521); no statement about large `p`.
- 40 curves per cell: trials are not independent draws of curves.
- JKSST’s `C0` is not computed (see PROTOCOL.md); the test is of the exponent.
- The collinear (line) mechanism means most successes need the *integer lifts* to be collinear, a property of the chosen lifts, not of the curve; the 13 non-line successes at `p >= 127` are too few to say anything about deeper dependence.
- S3: raw rows are not in S3 (no credential); they are committed gzipped and recorded by sha256 in the bench record.

Reproduce

`gunzip data/rows-p*.jsonl.gz` then `python3 analyze.py data` (writes `data/results.json` and `data/success_vs_p.svg`, copied here), then `python3 build_results.py` (tables and `dispersion.json`). The sweep itself: `python3 run_all.py suite/target/release/ca-icx data` from the repository root (trial seeds are derived from the master seed, so rows are reproducible independent of scheduling).